

```
In [1]: import pandas as pd

# Sample data
data = {
    'Student_ID': range(1, 11),
    'Name': ['John', 'Alice', 'Bob', 'Eve', 'Mike', 'Sara', 'Tom', 'Nina', 'Leo', 'Anna'],
    'Age': [20, 21, 19, 22, 20, 21, 23, 19, 20, 21],
    'Department': ['CS', 'IT', 'CS', 'BCA', 'CS', 'IT', 'BCA', 'CS', 'IT', 'BCA'],
    'Marks': [85, 90, 78, 92, 60, 88, 55, 95, 80, 70],
    'Attendance (%)': [90, 95, 85, 88, 60, 92, 50, 98, 87, 72]
}

df = pd.DataFrame(data)
print("Original Dataset:\n", df)
```

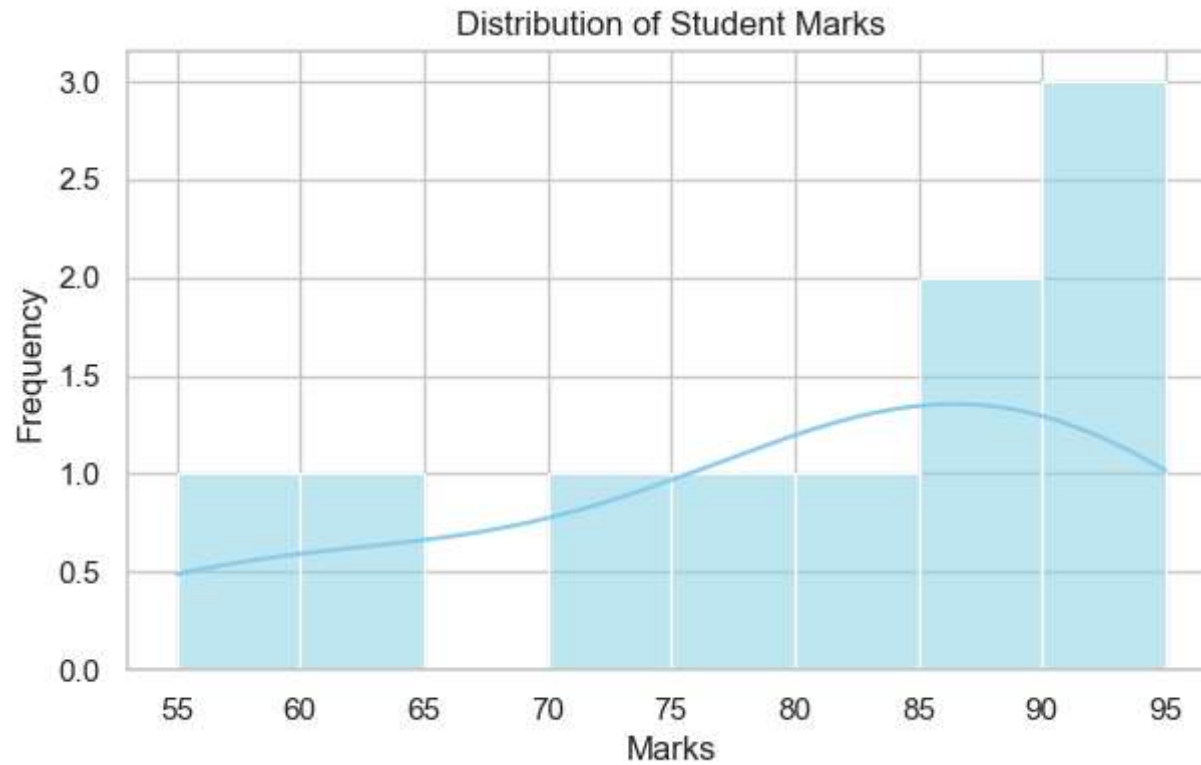
Original Dataset:

	Student_ID	Name	Age	Department	Marks	Attendance (%)
0	1	John	20	CS	85	90
1	2	Alice	21	IT	90	95
2	3	Bob	19	CS	78	85
3	4	Eve	22	BCA	92	88
4	5	Mike	20	CS	60	60
5	6	Sara	21	IT	88	92
6	7	Tom	23	BCA	55	50
7	8	Nina	19	CS	95	98
8	9	Leo	20	IT	80	87
9	10	Anna	21	BCA	70	72

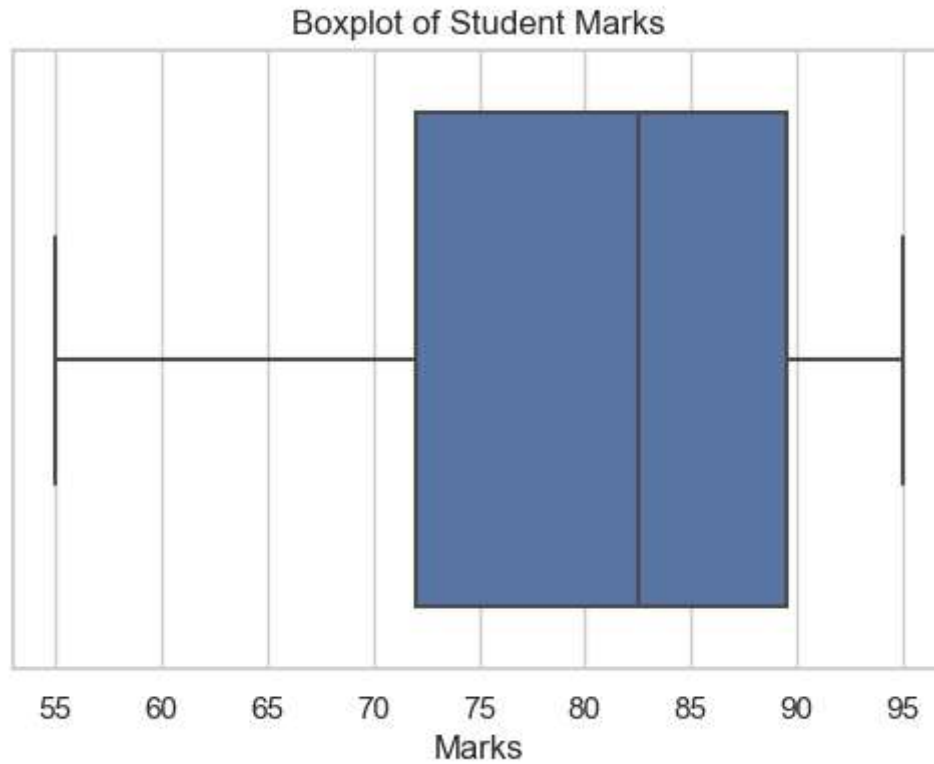
```
In [2]: import matplotlib.pyplot as plt
import seaborn as sns

# Set default style
sns.set(style='whitegrid')
```

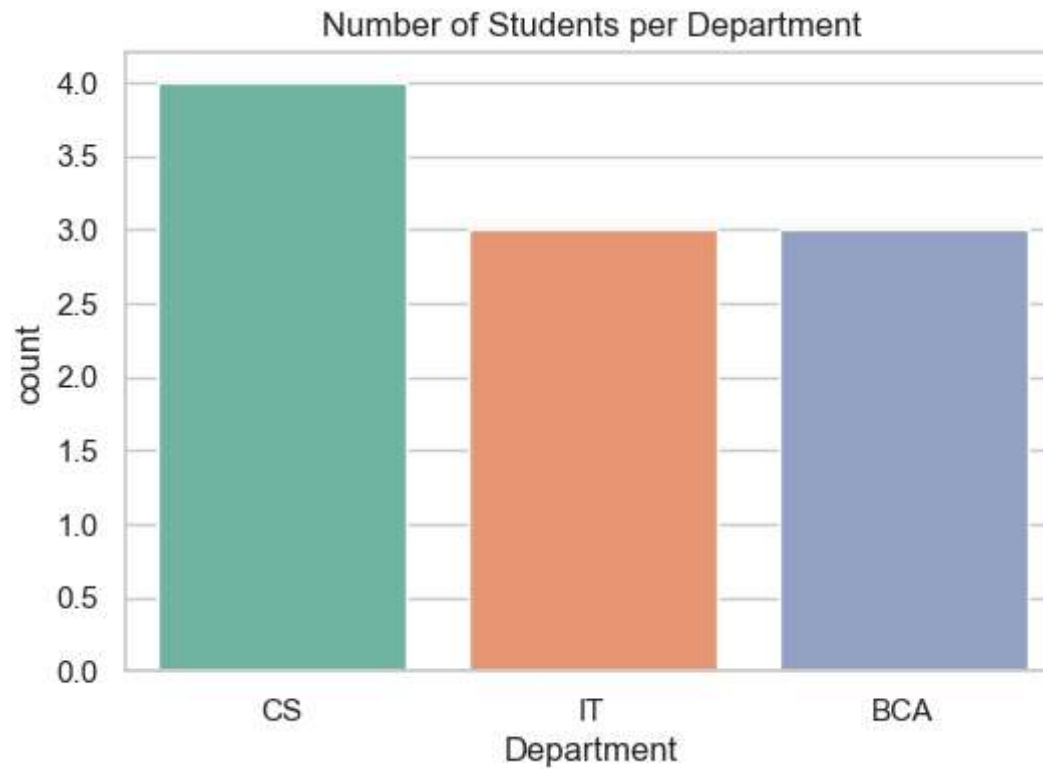
```
In [3]: plt.figure(figsize=(7, 4))  
sns.histplot(df['Marks'], bins=8, kde=True, color='skyblue')  
plt.title("Distribution of Student Marks")  
plt.xlabel("Marks")  
plt.ylabel("Frequency")  
plt.show()
```



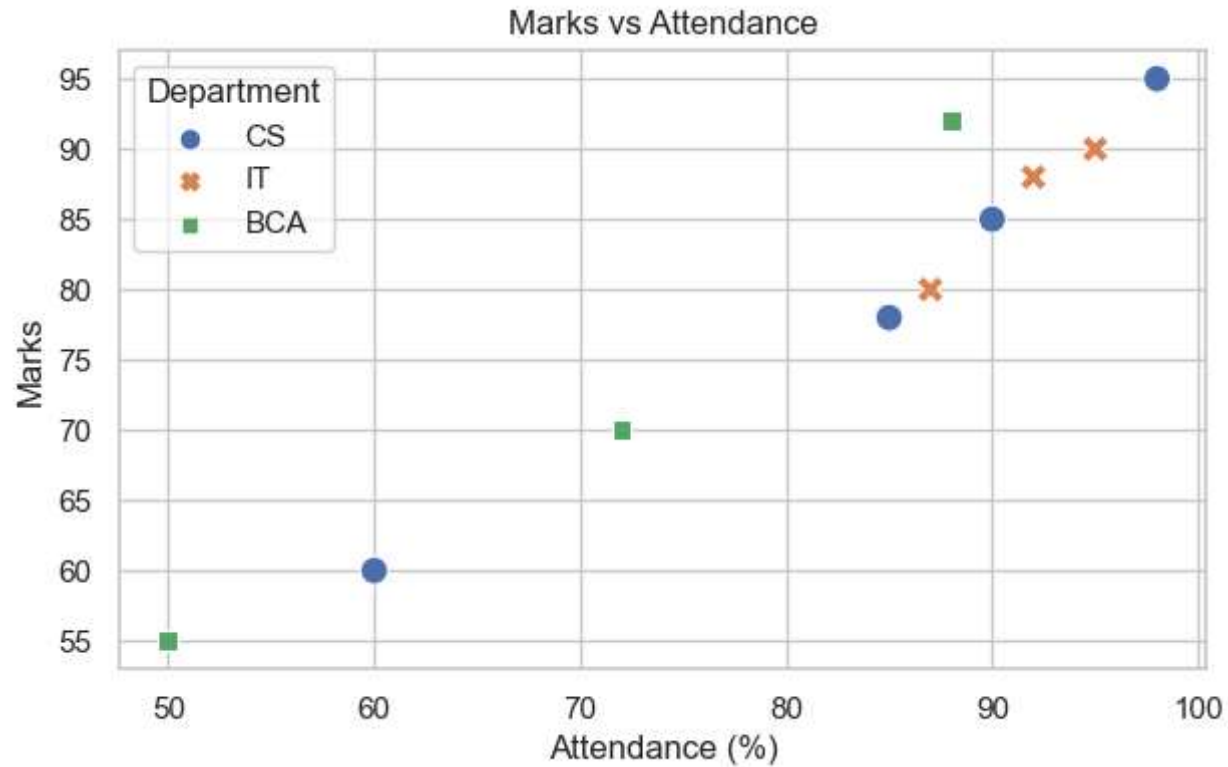
```
In [4]: plt.figure(figsize=(6, 4))  
sns.boxplot(x=df['Marks'])  
plt.title("Boxplot of Student Marks")  
plt.show()
```



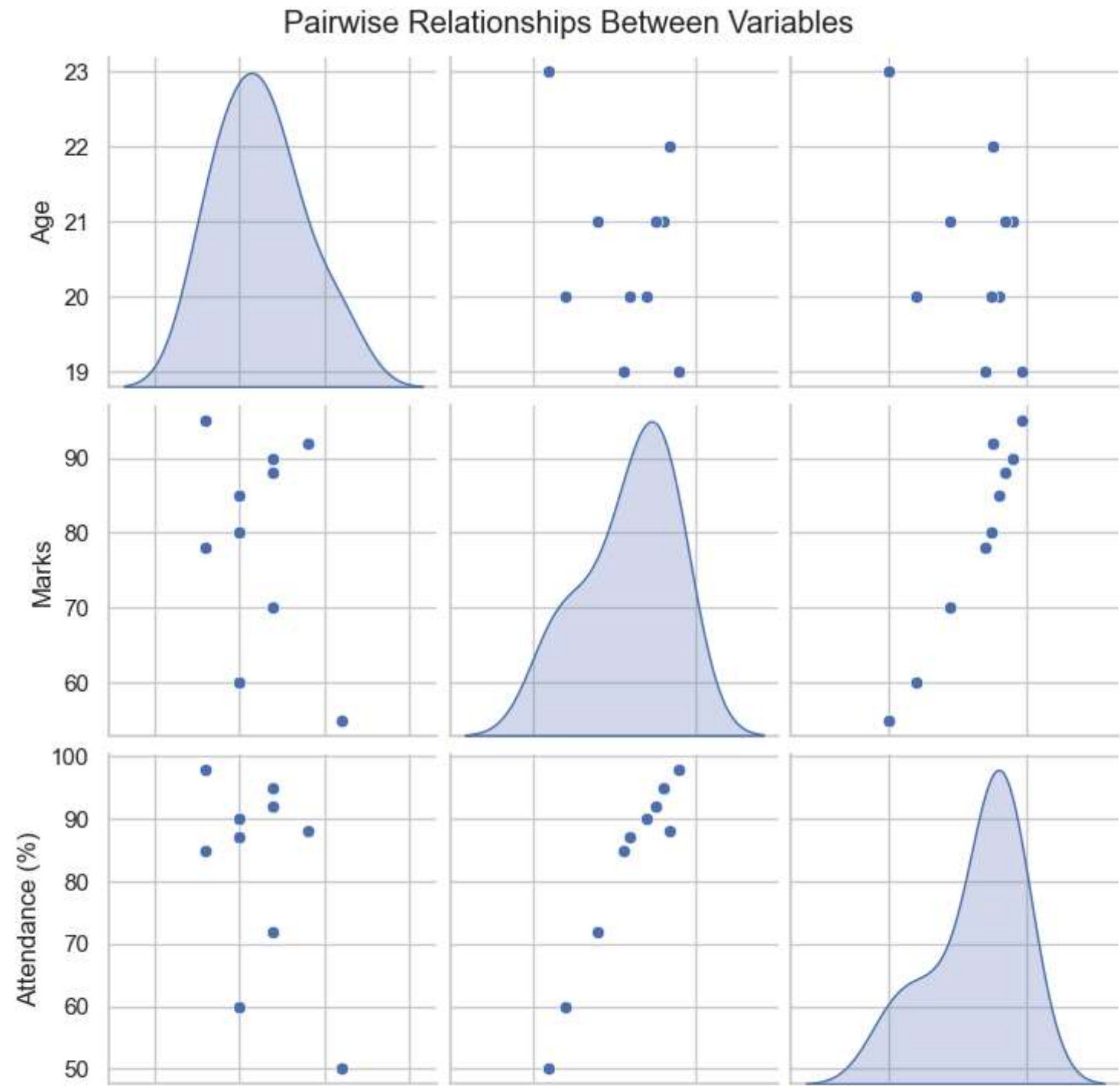
```
In [5]: plt.figure(figsize=(6, 4))  
sns.countplot(x='Department', data=df, palette='Set2')  
plt.title("Number of Students per Department")  
plt.show()
```



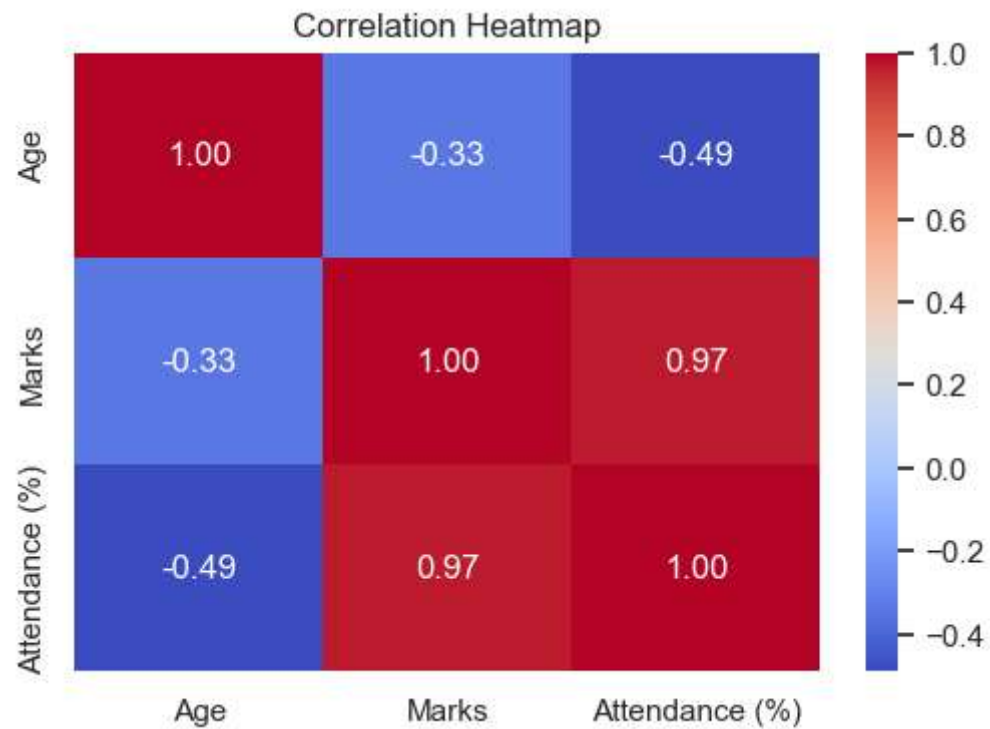
```
In [6]: plt.figure(figsize=(7, 4))  
sns.scatterplot(x='Attendance (%)', y='Marks', data=df, hue='Department', style='Department', s=100)  
plt.title("Marks vs Attendance")  
plt.show()
```



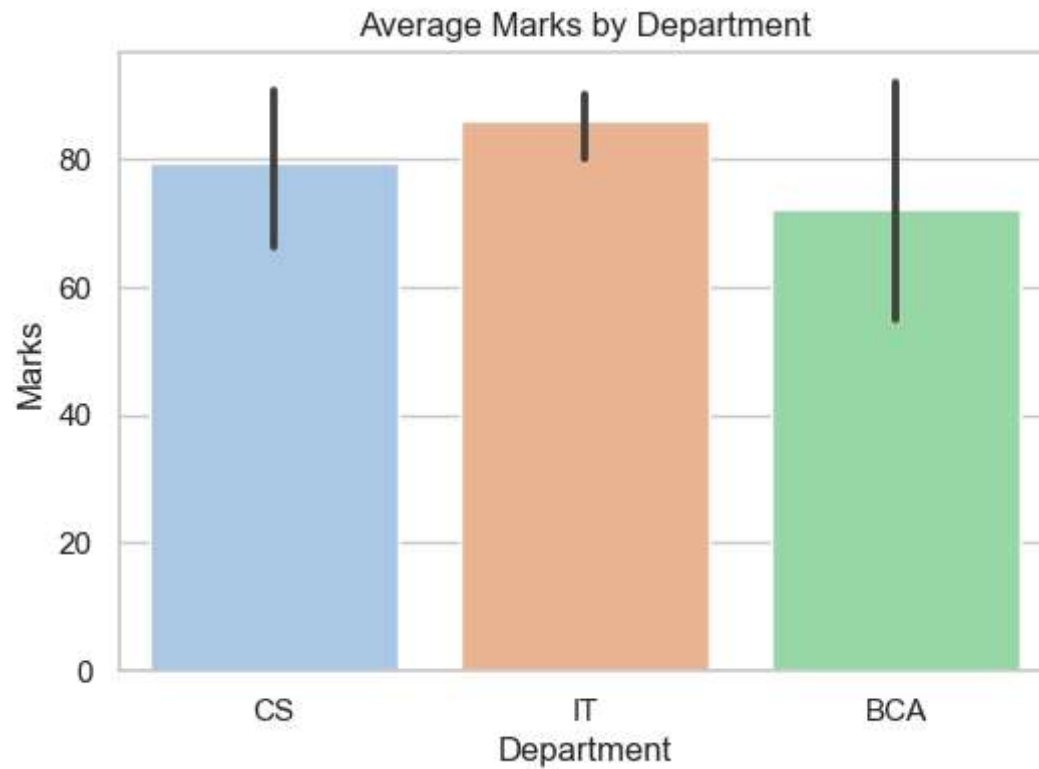
```
In [7]: sns.pairplot(df[['Age', 'Marks', 'Attendance (%)']], diag_kind='kde')  
plt.suptitle("Pairwise Relationships Between Variables", y=1.02)  
plt.show()
```

Age	Marks	Attendance (%)
17.5	50	50
20.0	100	100
22.5		
25.0		



```
In [9]: plt.figure(figsize=(6, 4))  
sns.barplot(x='Department', y='Marks', data=df, estimator='mean', palette='pastel')  
plt.title("Average Marks by Department")  
plt.show()
```



```
In [10]: print("\n✅ Insights:")
print("- The distribution of marks shows most students score between 70-90.")
print("- Outliers detected in Marks and Attendance for some students.")
print("- Attendance and Marks show a positive correlation.")
print("- CS department students generally score slightly higher on average.")
```

✅ Insights:

- The distribution of marks shows most students score between 70-90.
- Outliers detected in Marks and Attendance for some students.
- Attendance and Marks show a positive correlation.
- CS department students generally score slightly higher on average.

In []: